

SHREY AGARWAL

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EDUCATION

Northeastern University, Boston, MA

Expected Graduation: May 2027

Bachelor of Science in Computer Science and Business Administration

GPA: 3.3

Relevant Coursework: Algorithms & Data Structures, Probability & Statistics, Calculus II, Computational Methods in Finance

Markets & Trading: IMC Prosperity Trading Challenge — Top 8% of 18,000+ teams, Bloomberg Market Concepts, Bloomberg

Finance Fundamentals, Akuna Options 101, CME Group Introduction to Futures, CME Group Introduction to Oil

TECHNICAL SKILLS

Quantitative & Data: Python (Pandas, NumPy, Statsmodels), SQL (PostgreSQL), Excel (Advanced), C/C++ fundamentals

Software & Infrastructure: ETL Pipelines, REST APIs, AWS, Docker, CI/CD, Git, Java, Agile

EXPERIENCE

Quantitative Trader, Prediction Markets Fund | Moreton Capital Partners

April 2026 – Present

- Traded a \$2K allocation within a \$100K prediction-markets fund focused on event-driven opportunities across crypto, finance, and commodities markets
- Developed probability-based trading research using implied probabilities, probability-sum violations, spreads, market depth, liquidity, and time-to-resolution to identify executable mispricings
- Built and tested strategy logic for spread capture, momentum signals, risk-adjusted entry thresholds, slippage filters, and near-resolution trade avoidance
- Designed live-capital risk controls using capped fractional Kelly sizing, bankroll limits, exposure caps, liquidity constraints, and trade-level P&L/risk tracking

Quant Analyst | Disrupt: The FinTech Initiative

January 2026 – Present

- Evaluated equity, ETF, and options strategies in Python using Sharpe, drawdowns, volatility, and factor exposure metrics to assess risk-adjusted performance across market regimes
- Designed systematic backtests and scenario analyses to identify drivers of portfolio P&L, exposure, and capital efficiency across changing market conditions
- Engineered data workflows to clean and structure price, fundamental, and rates datasets, improving reliability of risk and performance analytics for quantitative research
- Summarized derivatives pricing, volatility, and exposure analysis into decision-oriented research for portfolio discussions

Software Developer Co-op | Northeastern University – College of Engineering

July 2025 – December 2025

- Built automated Python and SQL data pipelines processing 40,000+ records with validation, logging, and multithreaded refreshes for internal reporting systems
- Designed PostgreSQL schemas and ETL workflows to improve data integrity and scalability across large internal datasets
- Developed Streamlit dashboards to monitor classification outputs, pipeline health, and data quality metrics for technical and non-technical stakeholders
- Implemented LLM-based relevance scoring and authored documentation to improve reproducibility, stakeholder handoff, and long-term pipeline reliability

PROJECTS & RESEARCH

Order Book Simulator for Futures Markets

Python

- Built a futures order-book simulator modeling market orders, limit orders, cancellations, bid-ask spreads, depth, queue priority, and execution slippage across 3 core order types
- Simulated 10,000+ order events across 50+ liquidity and order-flow scenarios to analyze how spread, depth, and queue position affect realized execution price
- Quantified execution quality using fill probability, average slippage, adverse selection risk, and price impact metrics across 5+ market-depth levels
- Designed strategy-entry tests to evaluate how execution costs changed expected P&L, helping identify risk-adjusted entry thresholds under different liquidity conditions

Options Volatility Surface & Pricing Toolkit

Python, Polars, Streamlit, QuantLib

- Built a derivatives research toolkit to estimate implied volatility from option prices using Black-Scholes-style models
- Constructed interactive strike-maturity volatility surfaces to analyze volatility smiles and skews across market regimes
- Compared implied volatility against realized volatility through scenario and regime analysis to evaluate market sentiment, volatility risk, and potential mispricings
- Accelerated option-chain analysis using Polars and multithreading, improving research workflow speed across large derivatives datasets